

What the Brain Does

The brain is the body's control center. It coordinates the basic functions required for survival, controls body movements, and processes sensory data. However, it also encodes a lifetime of memories and creates consciousness, imagination, and our sense of self.

DO BRAINS FEEL PAIN?

Despite the fact that it registers pain from around the body, brain tissue has no pain receptors and cannot feel pain itself.

The physical brain

At the largest scale, the human brain appears as a firm, pink-gray solid. It is made mostly from fats (about 60 percent) and has a density just a little greater than that of water. However, neuroscientists, the people who study the form and function of the brain, see the organ as being constituted from more than 300 separate, although highly interconnected, regions. On a much smaller scale, the brain is made from approximately 160 billion cells, half of which are neurons, or nerve cells, and about half are glia, or support cells of one kind or another (see pp.20–21).



Weight

On average, an adult human brain weighs 2.6–3.1 lb (1.2–1.4 kg), which is approximately 2 percent of total body weight.



Fat

The brain's dry weight is 60 percent fat. Much of this fat is present as sheaths coating the connections between neurons.



Water

The brain is 73 percent water, while the body as a whole is closer to 60 percent. The average brain contains around 35 fl oz (1 liter) of water.



Volume

The average volume of a human brain ranges from 69 to 77 cubic in (1,130 to 1,260 cubic cm), although the volume decreases with age.



Gray matter

About 40 percent of the brain's tissue is gray matter, which is tightly packed nerve-cell bodies.



White matter

Around 60 percent of the brain's tissue is white matter. This is made from long, wirelike extensions of nerve cells covered in sheaths of fat.

LEFT BRAIN VS. RIGHT BRAIN

It is often claimed that one side, or hemisphere, of the brain dominates the other—and that this has an impact on someone's personality. For example, it is sometimes said that logical people use their left brain hemisphere, while artistic (and less logical) people rely on the right side. However, this is an extreme oversimplification. While it is true that the hemispheres are not identical in function—for example, the speech centers are normally on the left—most healthy mental tasks deploy regions on both sides of the brain at the same time.

